Symbolic Discovery of Explicit and Auditable Spectral Models for Land-Cover Mapping

Junwu Dong, Tao Pei, Ci Song, Xi Wang, Yahui Qi, Dayu Cheng, and Yanmin Jiang

***Abstract*—Remote sensing intelligence has long faced a fundamental tradeoff between model accuracy and interpretability. Deep learning models offer strong nonlinear representation capacity but remain intrinsically opaque, whereas traditional spectral indices, though physically interpretable, rely on fixed human-designed formulations that cannot adapt to complex or evolving observation scenarios. To address this gap, we propose a data-driven symbolic discovery framework that evolves explicit, task-adaptive spectral models from multispectral observations. The framework comprises two complementary paradigms. The symbolic transformer (ST) performs single-index discovery for threshold-based land-cover extraction using the directional Otsu separability fitness. The symbolic classifier (SC) performs synergistic index-set discovery for multiclass discrimination by jointly evolving calibrated class-score functions under multiclass hinge loss. Experiments on hyperspectral and multispectral benchmarks show competitive performance against conventional machine-learning, spectral deep learning, and symbolic-learning baselines. Across 12 hyperspectral datasets, SC ranked second in average Recall and *F*1-score among 14 evaluated methods. Compact ST expressions recover established index structures and reveal spectrally plausible relationships. Compact SC expressions likewise support physically informed inspection of band relationships, whereas more complex SC models retain explicit class scores and auditable interclass margins. Together, the two paradigms support compact spectral-index discovery and auditable multiclass decision modeling. Furthermore, the codes used in this study are available at https://github.com/djw-easy/symbolearn**

***Index Terms*—Genetic programming (GP), interpretable machine learning, land-cover classification, spectral indices discovery, symbolic learning.**

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I. Introduction

THE rapid deployment of new-generation Earth observation satellites, such as Sentinel-2 and Landsat-8/9, has brought remote sensing into an unprecedented era of data abundance [1]. The massive volume and high dimensionality of multispectral imagery provide a solid foundation for fine-scale monitoring of Earth system dynamics, while simultaneously imposing greater demands on intelligent image interpretation [2]. A key challenge is to fully exploit the information contained in these data while maintaining model reliability and interpretability—an issue critical in high-stakes applications such as disaster response, environmental change assessment, and sustainable resource management [3].

Balancing model performance and interpretability has long been a central problem in remote sensing analysis [4]. On the one hand, black-box models represented by deep learning achieve remarkable accuracy in land-cover classification through strong nonlinear representation capability. Recent efforts have further improved robustness under domain shifts through cross-domain adaptation and parameter-efficient spectral foundation models [5], [6]. However, the internal decision mechanisms of these models remain difficult to inspect, which limits their use in applications requiring transparency, traceability, and scientific understanding [7]. On the other hand, conventional spectral indices, such as the normalized difference vegetation index (NDVI) and normalized difference water index (NDWI), are compact analytical models with well-established physical interpretations [8]. Yet their manually designed and fixed mathematical forms often have limited adaptability to data variability, complex land-cover types, and changing observation conditions [9].

The traditional paradigm of spectral-index design relies heavily on expert knowledge and heuristic exploration, which faces two inherent bottlenecks [10]. First, the lack of optimality: general-purpose indices such as NDWI or enhanced vegetation index (EVI) are usually tailored for specific sensors or conditions and fail to maintain consistent performance across diverse environments [11], [12]. With the explosive growth of possible band combinations in modern high spectral sensors, manual exploration of optimal formulations has become impractical. Second, the lack of synergy: for multiclass classification, independently designed indices for different targets are often stacked as input features, ignoring interindex conflicts and redundancies, which leads to inconsistent or suboptimal discrimination [13].

Meanwhile, advances in spectral resolution enable more subtle differences between surfaces, while improved spa-

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tial resolution can reduce mixed-pixel effects and enhance class separability [14]. However, the conventional "manually designed, statically applied" paradigm cannot fully exploit this increasing spectral richness, creating a gap between data availability and interpretable modeling capability [15]. This motivates a data-driven mechanism that can discover discriminative spectral representations while keeping the resulting decision process and its structural complexity explicitly inspectable.

To address these challenges, this study draws inspiration from symbolic regression (SR) in explainable machine learning and proposes a unified symbolic spectral modeling framework. The framework evolves explicit and data-adaptive expressions directly from multispectral observations. The framework extends symbolic terminals from discrete spectral bands to continuous spectral aggregation features through a dynamic spectral aggregation mechanism. It also adopts complexity-based sampling to initialize structurally diverse expressions and supports the coordinated evolution of multiple expressions to capture complementary discriminative patterns. Within this framework, two complementary paradigms are developed: 1) symbolic transformer (ST) discovers compact single indices for automated threshold-based target extraction using an Otsu-based separability fitness and 2) symbolic classifier (SC) jointly evolves class-specific symbolic discriminant functions for multiclass classification through a hierarchical evolutionary mechanism. The main contributions include the following.

1) We propose a unified symbolic discovery framework that equips genetic programming (GP) with dynamic spectral aggregation terminals (DSATs) and node-count-based initialization for task-oriented remote sensing modeling.
2) We design an Otsu-inspired fitness function for automated threshold segmentation, enabling direct evolution of indices with strong target–background separability.
3) We develop a symbolic multiclass classifier that jointly evolves calibrated class-specific discriminant functions under a multiclass hinge loss, yielding an analytically specified and auditable multiclass decision procedure.

The remainder of this article is organized as follows. Section II reviews the related work on feature modeling and symbolic learning. Section III describes the proposed symbolic discovery framework for land-cover classification, including the GP components and optimization strategies. Section IV presents the experimental setup, datasets, and comparative results. Finally, Section V concludes this article and discusses limitations and future directions.

II. Related Work

A. Evolution of Remote Sensing Feature Modeling

Spectral indices have long been a classic, effective feature construction method in remote sensing, and a cornerstone of interpretable modeling for intelligent image understanding [16]. Representative indices such as NDVI and NDWI are widely used for large-scale environmental monitoring and targeted land-cover detection [8], [17]. Typically derived from expert knowledge, these indices combine two or more spectral bands through simple algebraic operations to enhance target–background contrast. Their compact mathematical form and clear physical meaning make them particularly suitable for threshold-based extraction and classification.

However, this expert-driven design paradigm faces increasing challenges in the era of high-dimensional remote sensing data. With the emergence of hyperspectral imagery, the number of possible band combinations has grown exponentially, making exhaustive manual design impractical and far from guaranteed to be optimal [18]. In addition, fixed-form indices often show limited robustness under varying atmospheric conditions, sensor noise, and seasonal changes [19]. More importantly, simple linear combinations or ratio-based formulations are often inadequate for capturing the complex and nonlinear spectral responses of heterogeneous land-cover types, such as urban mosaics or mixed vegetation [10].

To address these limitations, remote sensing research has increasingly shifted toward data-driven machine learning methods [20]. Algorithms such as support vector machines (SVMs) [21] and random forests (RFs) [22] can automatically learn nonlinear decision rules from raw spectral bands or handcrafted features, leading to substantial improvements in classification and segmentation accuracy [23]. More recently, deep learning models, including CNNs and Transformers, have achieved state-of-the-art performance through end-to-end representation learning [24]. Nevertheless, the decision processes of these models remain largely opaque and lack explicit physical interpretation. This black-box nature restricts their use in mission-critical scenarios that require transparency, auditability, and trustworthiness, such as disaster response, environmental regulation, and policy support. Therefore, a key challenge in modern remote sensing feature modeling is to achieve both strong discriminative ability and high interpretability.

B. Interpretable Modeling Based on SR

In recent years, SR has emerged as a promising approach for interpretable scientific modeling and knowledge discovery [25], [26]. Unlike conventional regression methods that assume a predefined functional form, SR automatically searches for both the structure and parameters of mathematical expressions using heuristic optimization algorithms such as GP [27]. The resulting formulas can be compact and directly inspectable; they support physical interpretation when their structure and variables are consistent with independent domain knowledge. Owing to this dual potential, SR has been used not only to rediscover fundamental physical laws [28], but also to uncover new empirical relationships in fields such as hydrology [29] and atmospheric modeling [30].

In remote sensing, several studies have explored the use of SR for the automatic construction of spectral indices [31]. However, existing efforts still face several limitations. First, many methods predefine the structural form of the index, such as normalized differences, and optimize only the numerical coefficients, thereby restricting structural diversity and limiting the discovery of novel formulations [32]. Second, most approaches evolve a single index independently, overlooking

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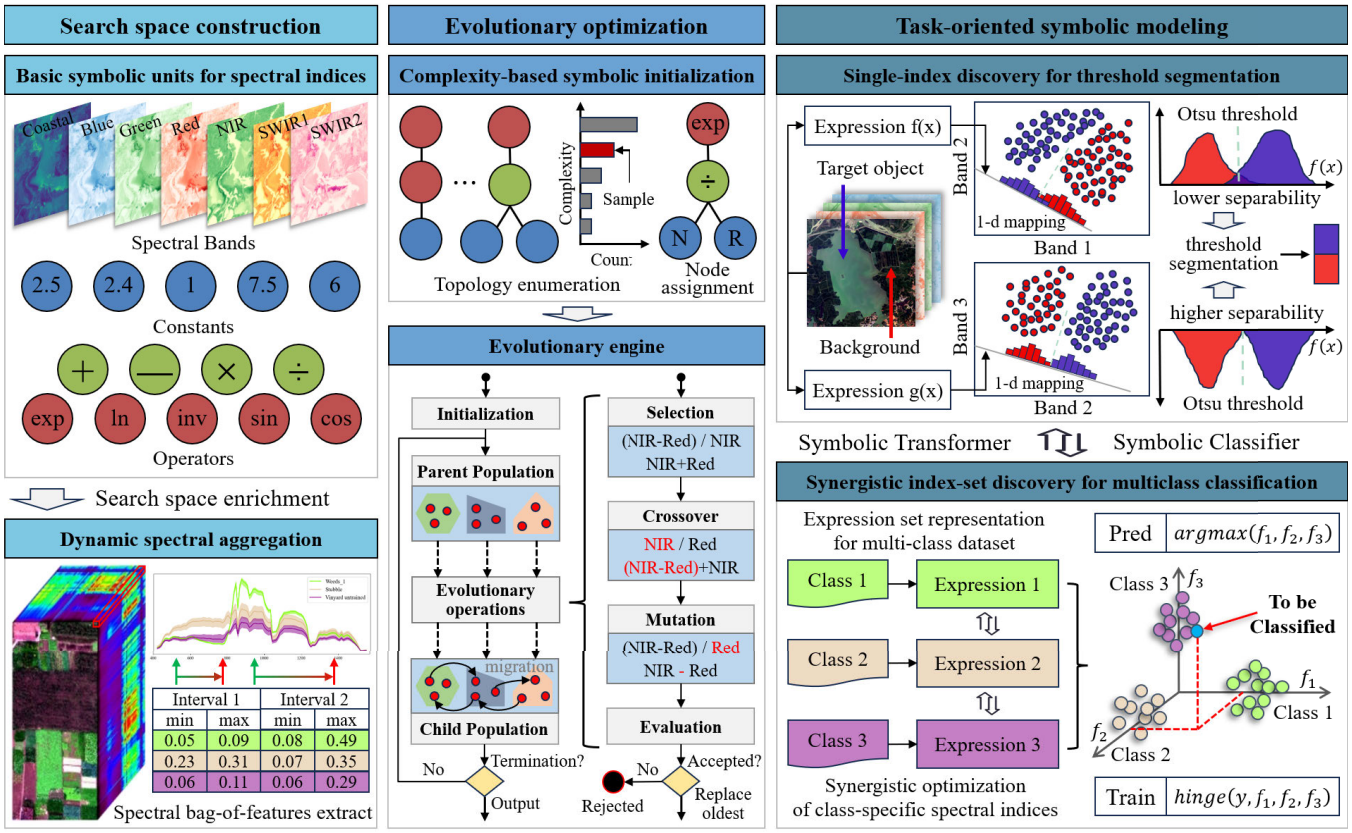


Fig. 1. Overview of the proposed unified framework for explicit symbolic spectral modeling.

the synergy and complementarity required for multiclass tasks, which may lead to feature redundancy or discriminative conflicts [33]. Third, insufficient complexity control often results in overly long expressions, weakening interpretability and reducing generalization ability [34].

Overall, current studies remain largely focused on single regression-style indices, and few have developed an end-to-end, task-oriented symbolic classification framework. For multiclass land-cover mapping, a more desirable strategy is to co-evolve a set of class-specific discriminative expressions under a unified objective, so as to ensure mutual complementarity and global consistency. Such a design moves beyond the traditional practice of stacking independently designed indices [35]. Therefore, integrating SR into remote sensing classification through threshold-oriented single-index discovery and end-to-end synergistic index-set discovery constitutes a largely unexplored research direction.

III. Methodology

This section describes the overall framework and key components of the proposed approach for explicit symbolic spectral modeling. As illustrated in Fig. 1, the framework comprises three stages: search-space construction, evolutionary optimization, and task-oriented symbolic modeling. Given multispectral observations and corresponding land-cover labels, the method constructs a symbolic search space from spectral variables, dynamic spectral aggregation features, and mathematical operators. A GP engine then searches this space to evolve symbolic expressions according to task-specific fitness functions.

The framework supports two complementary task-oriented paradigms. ST evolves a compact spectral index for one target land-cover class and maps each spectrum to a scalar response for threshold-based segmentation. SC jointly evolves an index set of class-specific symbolic expressions. Within this set, each index acts as a scalar class-score function, and the calibrated scores are compared to obtain the multiclass prediction. Thus, "index set" denotes a coordinated set of spectral score functions rather than independently thresholded physical indices.

To clarify the scope of interpretation for these paradigms, we distinguish three properties. *Syntactic transparency* denotes access to the analytical model form and the specified score-comparison procedure without hidden representations. *Structural complexity* describes expression size and composition. We characterize the structural aspect of formula readability using node count and operator types, which provide objective and reproducible measures of expression length and composition. *Physical interpretability* is a stronger, domain-specific property and is considered when a compact expression and its spectral variables are supported by independent knowledge of absorption, reflection, or scattering behavior. This criterion can apply to compact expressions discovered by either ST or SC. Complex SC expressions remain explicit and auditable, but their physical interpretation requires additional domain evidence.

A. GP-Based Symbolic Discovery Engine

All symbolic discovery tasks are implemented within a unified GP engine. GP evolves a population of candidate symbolic expressions through selection, crossover, and mutation to

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optimize a fitness function [27]. To balance exploration and convergence, we employ an island-model architecture in which multiple subpopulations evolve independently and periodically exchange elite individuals. We next describe the expression encoding and complexity-based initialization strategies used to construct a search space suitable for remote sensing spectral data.

1) Expression Encoding and Complexity Control: We adopt tree-based encoding because it naturally represents the hierarchical structure of symbolic spectral expressions and supports standard genetic operators. Each candidate expression is represented as a tree consisting of function nodes and terminal nodes. Function nodes denote mathematical operators, such as $+$, $-$, $\times$, and $\div$, whereas terminal nodes denote input spectral bands, constants, or DSATs. A DSAT aggregates reflectance over a continuous spectral interval $[b_s, b_e]$ using a statistical operator, such as mean, maximum, or minimum. For example, mean(b_4, b_{10}) denotes the mean reflectance from bands 4–10. The aggregation operator and interval boundaries are initialized randomly and subsequently evolved, enabling adaptive exploration of discriminative spectral regions.

Expression complexity is defined as the total number of nodes in its tree. It is constrained during evolution to limit code bloat and computational cost. In this study, node count and operator composition provide reproducible structural indicators of formula length and inspectability.

2) Complexity-Based Population Initialization: A diverse initial population is important for effective evolutionary search. Conventional Grow and Full initialization methods control tree depth, but depth does not accurately represent expression complexity: trees with the same depth can contain substantially different numbers of nodes depending on operator arity [36]. We therefore initialize expressions by directly controlling their node counts. The procedure consists of three steps.

1) *Counting Valid Topologies:* For each admissible complexity c, the number of valid tree topologies is recursively enumerated under the function-set and terminal-set constraints [Fig. 2(a) and (b)].
2) *Complexity Sampling:* A target complexity c^* is sampled from a discrete distribution proportional to the number of valid topologies at each complexity. A valid topology with c^* nodes is then selected randomly [Fig. 2(c)].
3) *Node Assignment:* Function nodes are assigned operators with compatible arities, whereas terminal nodes are assigned spectral bands, constants, or DSATs [Fig. 2(d)].

By sampling valid structures across the admissible complexity range, this strategy produces a structurally diverse initial population and helps reduce premature convergence.

B. Single-Index Discovery for Threshold Segmentation

Spectral indices combine spectral reflectances to enhance the contrast between a target land-cover type and its background, thereby projecting a high-dimensional spectral observation into a 1-D response. Let $\mathbf{X} \in \mathbb{R}^{N\times B}$ denote N pixel spectra with B spectral bands, and let $\mathbf{x} \in \mathbb{R}^B$ denote one pixel spectrum. ST evolves a scalar symbolic expression

$$f : \mathbb{R}^B \to \mathbb{R}, \qquad z = f(\mathbf{x}) \tag{1}$$

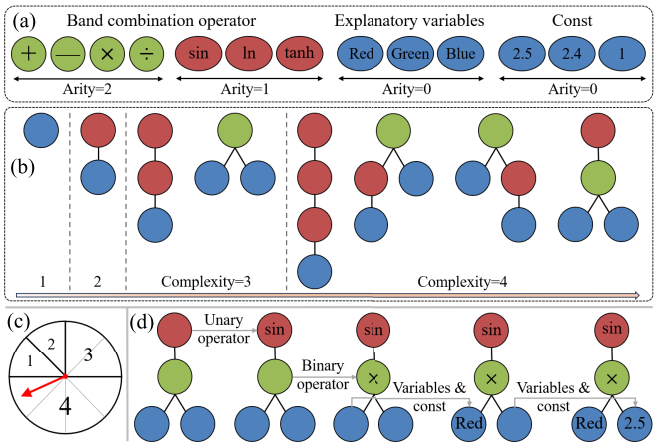


Fig. 2. Initialization strategy based on complexity distribution sampling. (a) Function and terminal sets. (b) Valid tree topologies by complexity. (c) Complexity-weighted topology sampling. (d) Assignment of operators and terminals to internal and leaf nodes, respectively.

where z is the scalar response. Evaluating f on all samples produces the response vector $\mathbf{z} \in \mathbb{R}^N$.

During GP training, the labeled target and background samples define two response groups. Let w_0 and w_1 denote their proportions and μ_0 and μ_1 their response means. We use the Otsu-inspired normalized between-group variance [37]

$$\eta = \frac{w_0 w_1 (\mu_0 - \mu_1)^2}{\mathrm{Var}(\mathbf{z})}, \qquad \eta \in [0, 1]. \tag{2}$$

Thus, the projection objective is label-conditioned and does not fit a Gaussian model or search for a threshold during evolution. Deployment-time Otsu threshold estimation is a separate operation applied to the response histogram of the fixed expression.

Because η is symmetric, it does not specify which response direction represents the target. We thus introduce a directional term based on the area under the receiver operating characteristic (ROC) curve (AUC) [38]

$$d = \mathrm{AUC}\left(\mathbf{z}_{\text{target}}, \mathbf{z}_{\text{background}}\right) \tag{3}$$

where $\mathbf{z}_{\text{target}}$ and $\mathbf{z}_{\text{background}}$ are the scalar-response vectors for target and background samples, respectively. Here, $d > 0.5$ indicates that target samples tend to produce larger responses than background samples. Defining $S_{\text{dir}} = 2(d - 0.5)$, the directional Otsu separability fitness is

$$F = 2\eta\,(d - 0.5), \qquad F \in [-1, 1]. \tag{4}$$

The multiplicative form makes the nonnegative separability η gate the signed directional term, so $F = 0$ when either $\eta = 0$ or $d = 0.5$; the factor 2 rescales the centered AUC to $[-1, 1]$.

Otsu thresholding is nonparametric and does not require Gaussian or symmetric class distributions. Nevertheless, deployment with one global threshold requires the discovered 1-D response to admit a useful target–background partition. Strongly multimodal backgrounds or extremely rare targets can violate this condition. We evaluate these limitations through heterogeneous-background and target-prevalence stress tests in Supplementary Section S8.

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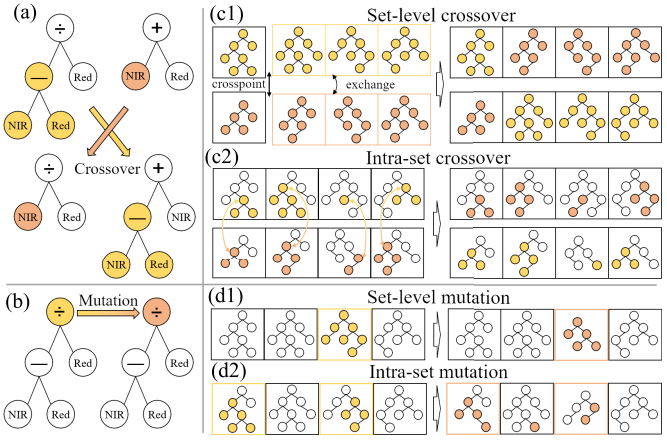


Fig. 3. Crossover and mutation operations for a single expression and an expression set. (a) Intra-expression subtree crossover. (b) Intra-expression mutation. (c.1) Set-level crossover between expression sets. (c.2) Subtree crossover between same-class expressions. (d.1) Set-level mutation of expression-set composition. (d.2) Intra-set mutation of individual expressions.

C. Synergistic Index-Set Discovery for Multiclass Classification

A single-index discovery method can be extended to multiclass classification using one-versus-rest models. However, directly extending the binary fitness through independently evolved one-versus-rest indices does not optimize the relative class scores that determine multiclass competition. We therefore propose the SC, which jointly evolves a set of class-specific symbolic discriminant functions under a multiclass objective.

1) Expression Set Encoding: Each individual is an expression set

$$\mathcal{S} = \{f_1, f_2, \ldots, f_C\} \tag{5}$$

where C is the number of classes and each scalar expression maps a pixel spectrum to a class-specific raw score

$$f_c : \mathbb{R}^B \to \mathbb{R}, \qquad c \in \{1, \ldots, C\}. \tag{6}$$

The set size is fixed during evolution. Here, "class-specific" denotes the score channel associated with one class in the multiclass competition, rather than an assumption that the class is spectrally unimodal.

2) Evolutionary Operators: We design a hierarchical evolutionary framework with set-level and intraset operations. Set-level crossover exchanges subsets of class-specific expressions between individuals, enabling class-wise knowledge transfer [Fig. 3(c.1)]. Intraset crossover exchanges subtrees between expressions for the same class [Figs. 3(a) and (c.2)]. Similarly, set-level mutation modifies the composition of an expression set [Fig. 3(d.1)], whereas intraset mutation modifies an individual expression tree [Figs. 3(b) and (d.2)].

3) Synergistic Optimization Mechanism: SC uses a multiclass hinge loss to jointly optimize all class-specific expressions [21]. For $\mathbf{x} \in \mathbb{R}^B$, the raw output of the cth expression is standardized using training-set statistics

$$\tilde{f}_c(\mathbf{x}) = \frac{f_c(\mathbf{x}) - \mu_c}{\sigma_c + \epsilon} \tag{7}$$

where μ_c and σ_c are the mean and standard deviation of $f_c(\mathbf{x})$ estimated from the training samples, and $\epsilon > 0$ is a numerical-stability constant. The training-set statistics are retained for validation and test samples.

The calibrated class-score vector and label prediction is

$$\mathbf{s}(\mathbf{x}) = \left[\tilde{f}_1(\mathbf{x}), \tilde{f}_2(\mathbf{x}), \ldots, \tilde{f}_C(\mathbf{x})\right]^{\mathsf{T}} \in \mathbb{R}^C \tag{8}$$

$$\hat{y}(\mathbf{x}) = \arg\max_{c \in \{1,\ldots,C\}} s_c(\mathbf{x}). \tag{9}$$

For a sample with true class c^*, the multiclass hinge loss is

$$\mathcal{L}(\mathbf{x}) = \frac{1}{C-1} \sum_{j \neq c^*} \max\left(0, 1 - \left[s_{c^*}(\mathbf{x}) - s_j(\mathbf{x})\right]\right). \tag{10}$$

The loss penalizes samples for which the true-class score fails to exceed competing scores by the desired unit margin, and is minimized via GP. Because the same calibrated scores are used for both optimization and prediction, simply rescaling a raw expression cannot create an artificial margin advantage.

IV. Experiments

This section evaluates the proposed symbolic discovery framework. We first describe the datasets, baselines, and evaluation metrics, followed by case studies of single-index discovery and synergistic index-set discovery. Comparative and ablation experiments then assess predictive generalization, structural complexity, and diagnostic transparency.

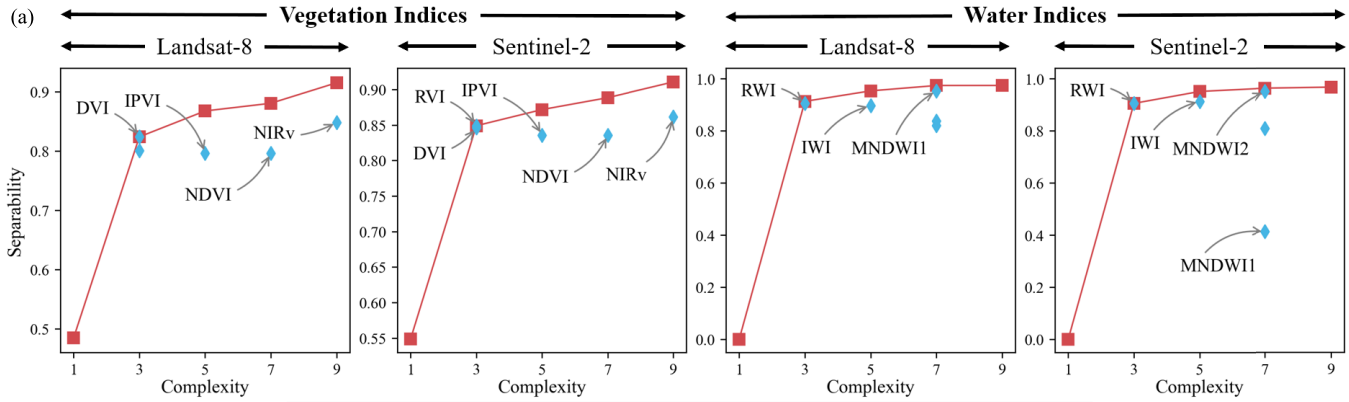
A. Experimental Setup

1) Datasets: This study uses three datasets to validate different paradigms. Dataset I (Poyang Lake) includes Landsat-8 operational land imagery (OLI) acquired on September 27, 2016 and Sentinel-2 Multispectral Instrument (MSI) imagery acquired on September 24, 2016, with binary water/nonwater and vegetation/nonvegetation labels, for evaluating the single-index discovery paradigm. Dataset II comprises two independent Landsat-8 scenes with contrasting land-cover compositions, Scene A (vegetation, water, built-up) and Scene B (vegetation, water, bare land), to assess the synergistic index-set discovery paradigm. Dataset III comprises 12 publicly available hyperspectral benchmark datasets with high dimensionality and diverse surface types. Their characteristics are summarized in Supplementary Table S1.

2) Baseline Models and Evaluation Metrics: To evaluate the performance of the proposed ST in spectral index discovery, we compared the automatically evolved indices with a set of widely used handcrafted spectral indices developed through prior empirical and physical reasoning. For vegetation extraction, the baseline indices included difference vegetation index (DVI) [39], ratio vegetation index (RVI), NDVI [8], infrared percentage vegetation index (IPVI) [40], and near-infrared reflectance of vegetation (NIRv) [41]. For water extraction, we considered ratio water index (RWI) [42], improved water index (IWI) [43], NDWI [17], MNDWI1 [44], and MNDWI2 [45], providing representative baselines for evaluating whether ST can match or exceed established manual formulations.

To evaluate the proposed SC, we selected comparison methods spanning conventional machine learning, deep spectral learning, and symbolic learning. The conventional baselines comprised k-nearest neighbors (kNNs) [46], decision tree (DT) [47], logistic regression (LR) [48], Naïve Bayes (NBs) [49],

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(b) **Symbolic Transformer Indices**

Complexity	Vegetation		Water	
	Landsat-8	Sentinel-2	Landsat-8	Sentinel-2
3	$\mathrm{NIR} - \mathrm{Red}$	$\frac{\mathrm{NIR}}{\mathrm{Red}}$	$\frac{\mathrm{Blue}}{\mathrm{SWIR1}}$	$\frac{\mathrm{Blue}}{\mathrm{SWIR2}}$
5	$\mathrm{NIR} - \mathrm{Red} - \mathrm{Red}$	$\mathrm{NIR} - \mathrm{Red} - \mathrm{Red}$	$\frac{\mathrm{Green}}{\mathrm{Green} + \mathrm{SWIR1}}$	$\frac{\mathrm{Green}}{\mathrm{Green} + \mathrm{SWIR2}}$
7	$\mathrm{NIR} - 2 * \mathrm{Red} + \mathrm{Blue}$	$\frac{\mathrm{NIR} - 2 * \mathrm{Red}}{\mathrm{Green}}$	$\frac{\mathrm{Blue}}{\mathrm{Green} + \mathrm{SWIR1}/\mathrm{Blue}}$	$\frac{\mathrm{Green}}{\mathrm{Green} + 2 * \mathrm{SWIR2}}$
9	$\frac{\mathrm{NIR}}{\mathrm{NIR} + \mathrm{Red}/\mathrm{NIR}} - \mathrm{NIR}$	$\frac{\mathrm{NIR}}{\mathrm{NIR} * \mathrm{NIR} * \mathrm{NIR} + \mathrm{Red}}$	$\frac{\mathrm{Blue}}{\mathrm{SWIR2} + \mathrm{G} + \mathrm{SWIR1}/\mathrm{B}}$	$\frac{\mathrm{Green}}{\mathrm{SWIR2} + \mathrm{Blue} * (\mathrm{G} + \mathrm{B})}$

(c) **Existing Indices**

Vegetation	Water
$\mathrm{DVI} = \mathrm{NIR} - \mathrm{Red}$	$\mathrm{RWI} = \mathrm{Blue}/\mathrm{SWIR2}$
$\mathrm{RVI} = \mathrm{NIR}/\mathrm{Red}$	$\mathrm{IWI} = \frac{\mathrm{Green}}{\mathrm{NIR} + \mathrm{SWIR2}}$
$\mathrm{IPVI} = \frac{\mathrm{NIR}}{\mathrm{NIR} + \mathrm{Red}}$	$\mathrm{NDWI} = \frac{\mathrm{Green} - \mathrm{NIR}}{\mathrm{Green} + \mathrm{NIR}}$
$\mathrm{NDVI} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red}}$	$\mathrm{MNDWI1} = \frac{\mathrm{Green} - \mathrm{SWIR1}}{\mathrm{Green} + \mathrm{SWIR1}}$
$\mathrm{NIRv} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red}} * \mathrm{NIR}$	$\mathrm{MNDWI2} = \frac{\mathrm{Green} - \mathrm{SWIR2}}{\mathrm{Green} + \mathrm{SWIR2}}$

Difference-based Ratio-based Normalized Difference-like Complex / Non-linear

Fig. 4. Analysis of spectral indices discovered by ST on the Poyang Lake dataset and their structural types. (a) Comparison of ST indices and existing indices on the "separability-complexity" Pareto frontier. (b) Expressions of the Pareto-optimal ST indices. (c) Expressions of the considered existing indices.

quadratic discriminant analysis (QDA) [50], SVM [21], RF [22], eXtreme gradient boosting (XGBoost) [51], light gradient boosting machine (LightGBM) [52], and multilayer perceptron (MLP) [53]. We further included three models: a GP-based symbolic model (M3GP) [54], a gradient-trained symbolic model (SymbolNet) [55], and the spectral-only deep model ODPA-CNN [56]. Performance was evaluated using Recall and $F1$-score. Hyperparameters for all baseline models were selected by grid search to ensure a fair comparison.

3) Implementation Details and Parameter Settings: All experiments were implemented using our custom GP framework. The compact configuration used a maximum complexity of 9, 31 islands with 27 individuals each, ten generations, and 380 evolutionary operations per generation. Its function set was $\{+, -, \times, \div\}$, and its terminal set contained the original spectral bands. For accuracy-oriented SC evaluation, we used 100 generations, increased the maximum complexity to 25 and the evolutionary operations to 780, and additionally enabled constants, DSATs, and $\{\mathrm{sigmoid}, \tanh, \mathrm{softplus}\}$. We targeted 100 training samples per class and repeated each experiment ten times using independent spatially disjoint partitions and model seeds. All methods received only the original spectral vector of the center pixel; no spatial aggregation or neighborhood-derived feature was used. Detailed GP settings and the spatial sampling protocol are provided in the Supplementary Material. A targeted sensitivity analysis of island granularity and function-set composition for Pavia University, Salinas, and WHU-Hi-HongHu is presented in Supplementary Section S12, Fig. S8, and Tables S14–S15.

B. Automatic Discovery and Verification of Spectral Index for Threshold Segmentation

Fig. 4 offers a comprehensive comparison between ST-discovered indices and classical handcrafted indices from the perspectives of separability and structural complexity, thus enabling a rigorous evaluation of solution optimality rather than performance alone. Fig. 4(a) shows that separability-complexity Pareto frontiers across four settings (Landsat-8 versus Sentinel-2; vegetation versus water) place ST-derived indices on or dominating the optimal Pareto boundary. Thus, for any human-designed index, ST can yield equal or superior separability at comparable or lower complexity. For example, although NDVI (complexity 7) is often regarded as near-optimal for vegetation, ST discovers a complexity-5 index that surpasses its separability, showing that conventional formulations may not be true upper bounds under data-adaptive optimization.

Fig. 4(b) visualizes structural evolution with increasing model complexity. At low complexity (1–3), ST reconstructs difference- or ratio-based forms, including NIR − Red, NIR/Red, and Blue/SWIR2, that resemble DVI, RVI, and RWI. At higher complexity (≥5), ST produces previously unreported but spectrally plausible combinations, including Green/(Green + SWIR1) and NIR −2× Red + Blue. These expressions recover band contrasts consistent with established vegetation and water responses, supporting structural rediscovery and providing physically plausible spectral relationships for further domain investigation.

Using Landsat-8 imagery, Fig. 5 qualitatively demonstrates the discriminative power and structural advantages of

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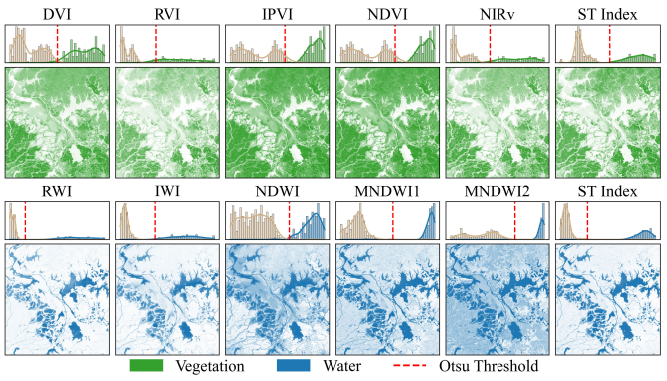


Fig. 5. Vegetation and water distributions obtained using Otsu thresholding on the Poyang Lake dataset.

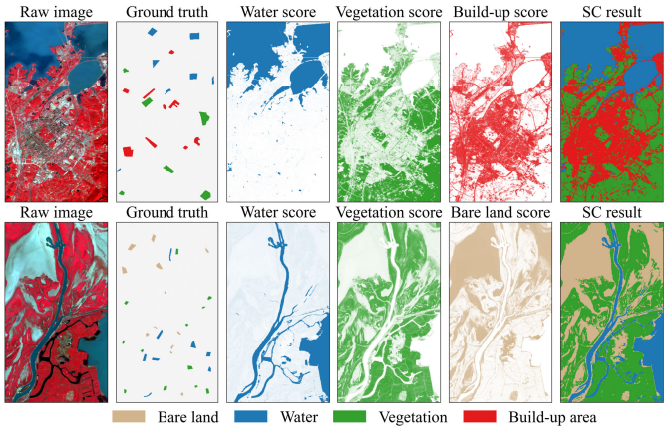


Fig. 6. Predicted class-score and classification maps of SC on the multiclass-L8 dataset.

ST-derived vegetation and water indices by comparing them with five established manual indices. The upper panels show each index's response distribution with the Otsu-derived optimal threshold, and the lower panels display the corresponding spatial response maps, illustrating separability across the study area. For vegetation, traditional indices exhibit substantial distributional overlap, hindering class separation and causing thresholding errors. In contrast, the ST index shows a bimodal distribution with well-separated peaks and an Otsu threshold located in the interpeak valley; the spatial map mirrors this with higher image contrast and sharper land-cover boundaries, surpassing traditional difference- or ratio-based indices. The ST water index likewise yields strong class decoupling: non-water pixels cluster in a narrow low-value range, water pixels form a high-value peak with a clear gap, and the spatial map suppresses land responses while sensitively delineating water boundaries, producing contiguous water extraction.

Overall, Fig. 5 indicates that ST can discover spectral indices with favorable statistical and spatial separability under the stated separability criterion, thereby improving visual class separation in threshold-based segmentation.

C. Decision-Behavior and Diagnostic Transparency Analysis of the Synergistic Index Set

To evaluate the adaptability and decision behavior of SC, we first compared its learned decision boundaries with those of nine baseline classifiers on six synthetic 2-D datasets. As shown in Fig. S3, SC produces smooth and well-organized decision boundaries that closely follow the underlying data manifolds. On linearly separable datasets, it converges to compact and simple boundaries, demonstrating its ability to represent concise functional forms. In more complex nonlinear scenarios, SC still constructs smooth, continuous, and topologically coherent boundaries that remain consistent with the intrinsic geometric structure of the data. Building on this proof-of-concept analysis, Fig. 6 examines SC on two real Landsat-8 scenes. In Scene A, the class-specific scores show complementary and spatially distinct activation: the water score responds to water bodies, whereas the vegetation and built-up scores delineate their respective surfaces. Unlike independently evolved ST indices (Fig. S2), which may activate simultaneously, SC jointly optimizes the complete index set under the multiclass hinge loss to produce a coherent multiclass decision. Tables S2 and S3 list the corresponding ST and SC indices. The compact SC expressions in Table S3 use standard Landsat-8 bands and simple algebraic operations, allowing their band relationships to be examined using spectral-domain knowledge. Scene B provides a more challenging vegetation–bare-land case. SC separates riverbanks and agricultural fields from tidal flats and exposed sediments, including transitional areas, under strong spectral ambiguity.

To further reveal the internal decision structure of SC, Fig. 7 presents class-wise score distributions, pairwise margins, and margin-induced confusion patterns on the Trento dataset. The diagonal panels show the distribution of each class-specific score $s_c(\mathbf{x})$, which maps an input spectrum to a 1-D symbolic response. Separation between the target-class distribution and the remaining classes reveals how each explicit expression supports class-specific decisions. The upper-triangular panels show the pairwise margins $s_i(\mathbf{x}) - s_j(\mathbf{x})$, whose sign indicates preference for class i or j. The associated AUC quantifies pairwise separability in this symbolic margin space. The lower-triangular panels show the decisions obtained from the sign of the same margin. Larger AUCs generally coincide with clearer separation and higher pairwise Recall and $F1$-score, as observed for buildings and woods. Smaller AUCs indicate greater overlap and lower performance, as observed for buildings and roads. Together, these panels identify difficult class pairs and relate their errors to narrow interclass margins or high intraclass variability.

Taken together, Figs. 6 and 7 show that SC organizes multiclass classification through an explicit set of coordinated symbolic scoring functions and their competitive margins. The class-wise scores show the response of each expression, while pairwise margins show how the score functions interact to produce the final decision. Unlike conventional black-box classifiers, which encode interclass competition in high-dimensional parameters, SC exposes class-oriented responses and interclass decision dynamics in an analytically accessible form. This makes the classifier predictive, structurally inspectable, and diagnostically transparent. Its explicit form supports analysis of class-discriminative spectral relationships without assigning a unique physical mechanism to every term in a complex expression.

We further audited intraclass spectral-mode coverage on Pavia University using independently defined and evaluated

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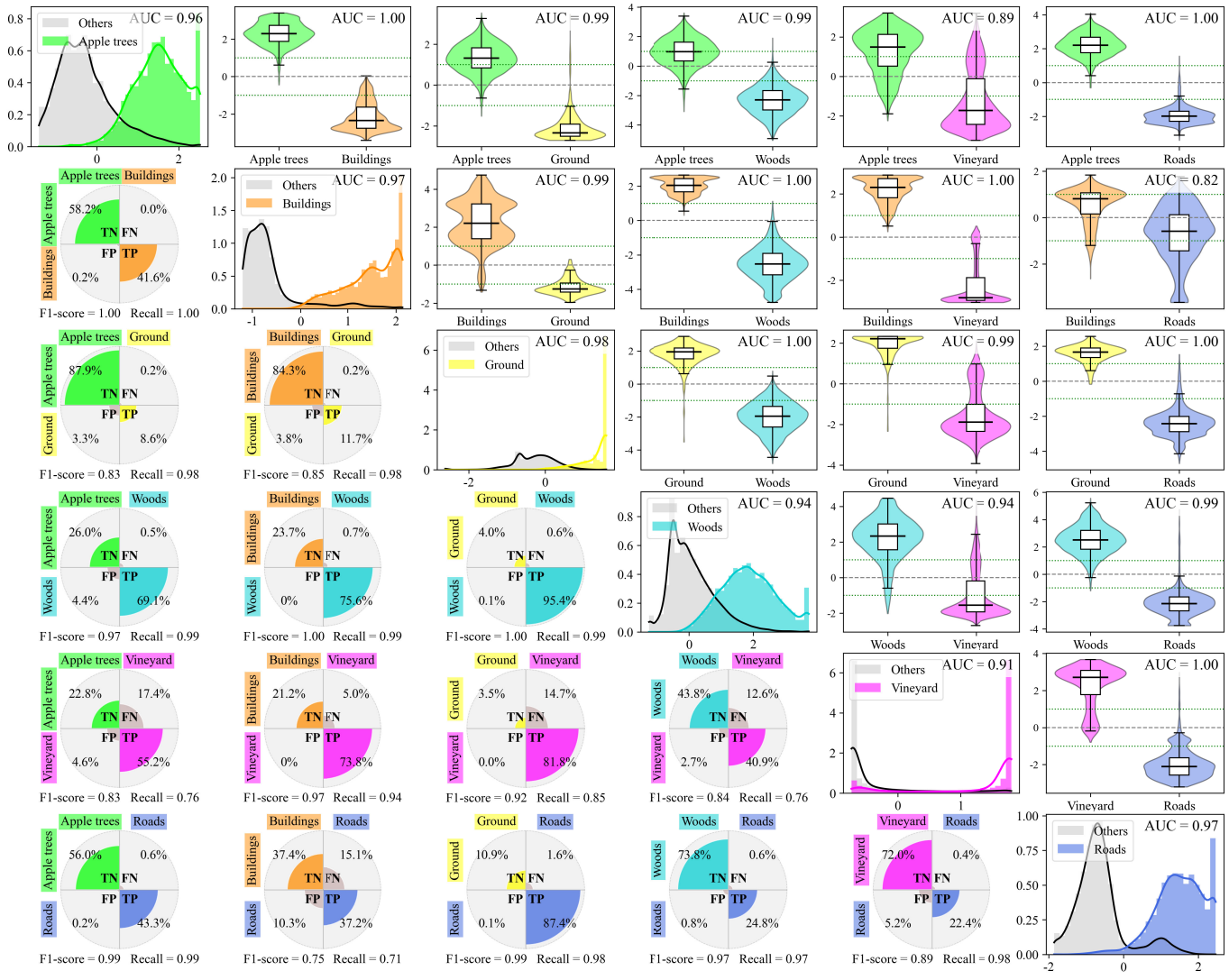


Fig. 7. SC decision-behavior matrix on the Trento dataset: class-wise responses, pairwise margins, and margin-induced confusion patterns.

modes that were not used for model training. Although SC achieved a median macroclass Recall of 0.743, its median mean class-wise worst-mode Recall was 0.447 (LightGBM: 0.463). The SC overall-to-worst-mode gap ranged from 0.016 for Painted metal sheets to 0.566 for Trees, showing that aggregate class Recall can conceal class-dependent mode failures. Full protocols and results are provided in Supplementary Section S9, Fig. S6, and Tables S8–S10.

D. Ablation Study and Quantitative Evaluation

This section evaluates SC through ablation and quantitative experiments. The ablation examines synergistic co-evolution, complexity-based initialization, and DSATs; the quantitative evaluation compares SC with conventional machine-learning, symbolic-learning, and spectral deep learning models on multiple hyperspectral benchmarks.

1) Ablation Study: The ablation was conducted on Pavia University, Salinas, and WHU-Hi-HongHu, which contain 9, 16, and 22 classes, respectively, and therefore represent progressively more challenging multiclass settings. The maximum permitted node count was varied over $\{5, 9, 13, 17, 21, 25, 31\}$. We compared four configurations: independently evolved ST combined in a one-versus-rest scheme, SC without complexity-based initialization (w/o init), SC without DSATs (w/o aggregation), and the complete SC. This design isolates the effects of synergistic co-evolution, initialization, and aggregation under matched complexity constraints.

Fig. 8 reports Recall across the three datasets. Although the benefit of additional nodes diminishes at high complexity, complete SC achieves the strongest or comparable Recall at most complexity levels. In contrast, the one-versus-rest ST baseline is consistently inferior, demonstrating the value of jointly optimizing class-specific expressions with the multiclass hinge loss. Removing either complexity-based initialization or DSATs generally reduces Recall, especially at higher node-count settings. These results support the complementary roles of the two proposed modules in improving symbolic search and multiclass discrimination.

We additionally examined the empirical optimization trajectory on the same three datasets using a fixed spatial partition and ten independent evolutionary seeds. Fig. 8(d) reports the median best-so-far hinge-loss fitness over 100 evolutionary generations, with the interquartile range shown as shading.

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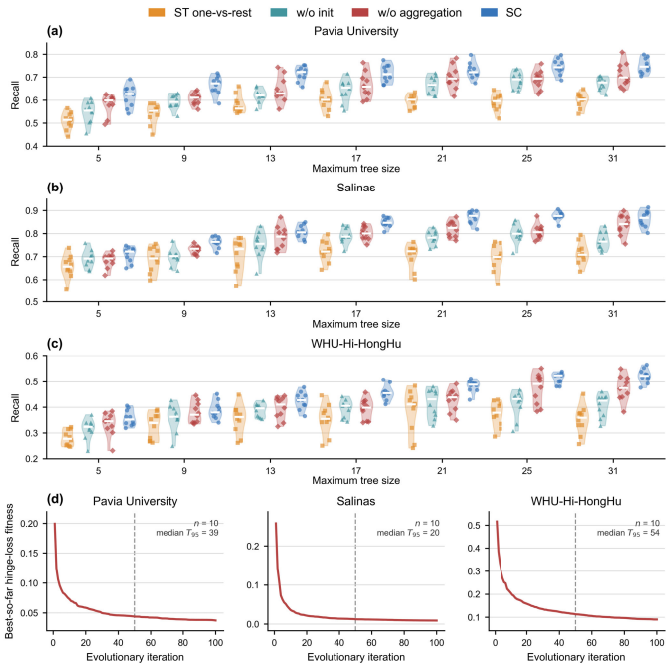


Fig. 8. Ablation and empirical convergence results. (a)–(c) Recall distributions for the four configurations across maximum tree sizes on Pavia University, Salinas, and WHU-Hi-HongHu, respectively. (d) Median best-so-far hinge-loss fitness over ten evolutionary seeds.

The median generation at which 95% of the improvement observed between generations 1 and 100 had been attained (T_{95}) was 39 for Pavia University, 20 for Salinas, and 54 for WHU-Hi-HongHu. After generation 50, the median remaining fractions of the observed improvement were 3.37%, 1.41%, and 5.60%, respectively. Together, the trajectories show rapid early fitness reduction and stable late-stage refinement across evolutionary seeds. Population-level distributions and the complete protocol are provided in Supplementary Section S11.

2) Quantitative Evaluation: Fig. 9 compares 14 methods across the 12 hyperspectral benchmarks. ODPA-CNN achieved the lowest average ranks for Recall and $F1$-score, with values of 2.0 and 2.33, respectively. SC ranked second overall, with average ranks of 2.50 for Recall and 2.67 for $F1$-score. Among the symbolic-learning baselines, SC substantially outperformed M3GP and SymbolNet, with Recall ranks of 9.83 and 7.25 and $F1$-score ranks of 9.92 and 6.58. For both metrics, SC also achieved better average ranks than the conventional kNN, DT, LR, NB, QDA, and SVM classifiers, as well as the RF, XGBoost, and LightGBM tree ensembles. These results place SC close to the spectral deep learning reference while outperforming the evaluated symbolic and conventional baselines and preserving an explicit symbolic decision structure. Each heatmap cell reports the mean over ten independent repetitions, and the orange bar length encodes the corresponding standard deviation. Complete mean $\pm$ std values are provided in Supplementary Tables S11–S12.

The Friedman test and Nemenyi analysis in Fig. 9(c) place SC within the leading group rather than establishing universal superiority over ODPA-CNN. SC nevertheless separates clearly from the lower-ranked symbolic baselines across the benchmark suite. Fig. 9(d) shows the method-specific training and testing Recall distributions, making the remaining generalization gaps visible. The learning curves in Fig. 9(e)–(g) show increasing Recall with additional training samples. SC remains competitive in the low-data regime and approaches a stable performance level as the training budget increases. Overall, the results support SC as a competitive spectral classifier when transparency and inspectable analytical structure are required.

Fig. 10 provides a qualitative comparison of the classification maps produced by different methods on the Salinas dataset. Overall, the proposed SC yields the map most consistent with the ground truth, accurately preserving the shapes of homogeneous regions, delineating narrow field boundaries, and effectively suppressing salt-and-pepper noise. In particular, SC exhibits superior separation of spectrally similar classes, including the different lettuce growth stages, fallow-related categories, and vineyard-related areas, where many competing methods tend to introduce local confusion or boundary leakage. These observations are consistent with the quantitative results and indicate that SC produces spatially coherent classification outputs while retaining an inspectable score-based decision procedure.

V. Conclusion

This article proposed a unified symbolic spectral modeling framework for remote sensing. It integrates ST for single-index discovery in threshold-based target extraction and SC for synergistic index-set discovery in multiclass classification. Both paradigms use a common GP engine with task-specific objectives: the directional Otsu separability fitness for binary extraction and multiclass hinge loss for multiclass discrimination.

Experiments on multispectral and hyperspectral datasets showed different performance–complexity tradeoffs for the two paradigms. ST discovered compact indices on or beyond the separability-complexity Pareto frontier defined by the evaluated handcrafted indices. These expressions recovered established structural patterns and generated spectrally plausible combinations for domain examination. SC evolved explicit class-specific score functions that jointly formed an auditable decision system. Across 12 hyperspectral benchmarks, SC ranked second in both Recall and $F1$-score among 14 conventional, deep spectral, and symbolic-learning methods. The ablation results also support the contributions of DSATs and complexity-based initialization to multiclass discrimination and symbolic search.

The framework provides complementary forms of model intelligibility. Compact expressions discovered by either ST or SC support structural inspection and can motivate physically grounded interpretation when their band relationships are supported by spectral knowledge. For higher complexity SC models, the principal interpretive value lies in explicit class scores and pairwise margins that support decision auditing and diagnostic analysis. ST remains sensitive to heterogeneous backgrounds and rare-target threshold bias, whereas SC may require mode-aware or multiexpression extensions when worst-mode robustness is important. The computational cost of SC grows with the evolutionary budget and the number of

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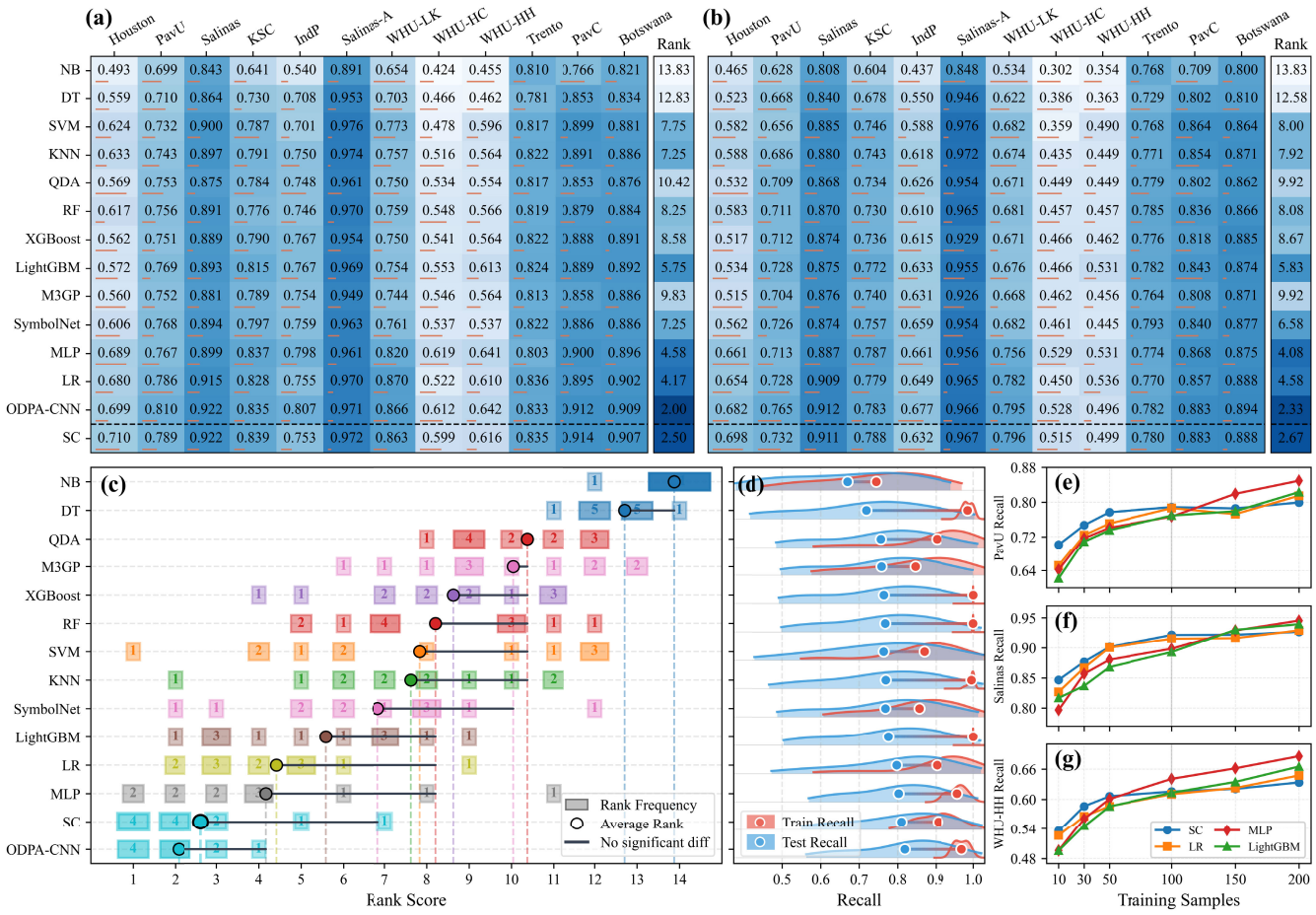


Fig. 9. Comprehensive quantitative evaluation across 12 benchmark datasets. (a) Recall and (b) $F1$-score; cell values are ten-run means, and orange bar lengths encode the corresponding standard deviations. (c) Rank frequencies, average ranks, and Nemenyi nonsignificant groups computed from the ten-run means. (d) Training and testing Recall distributions. (e)–(g) Recall trends with increasing training-set size.

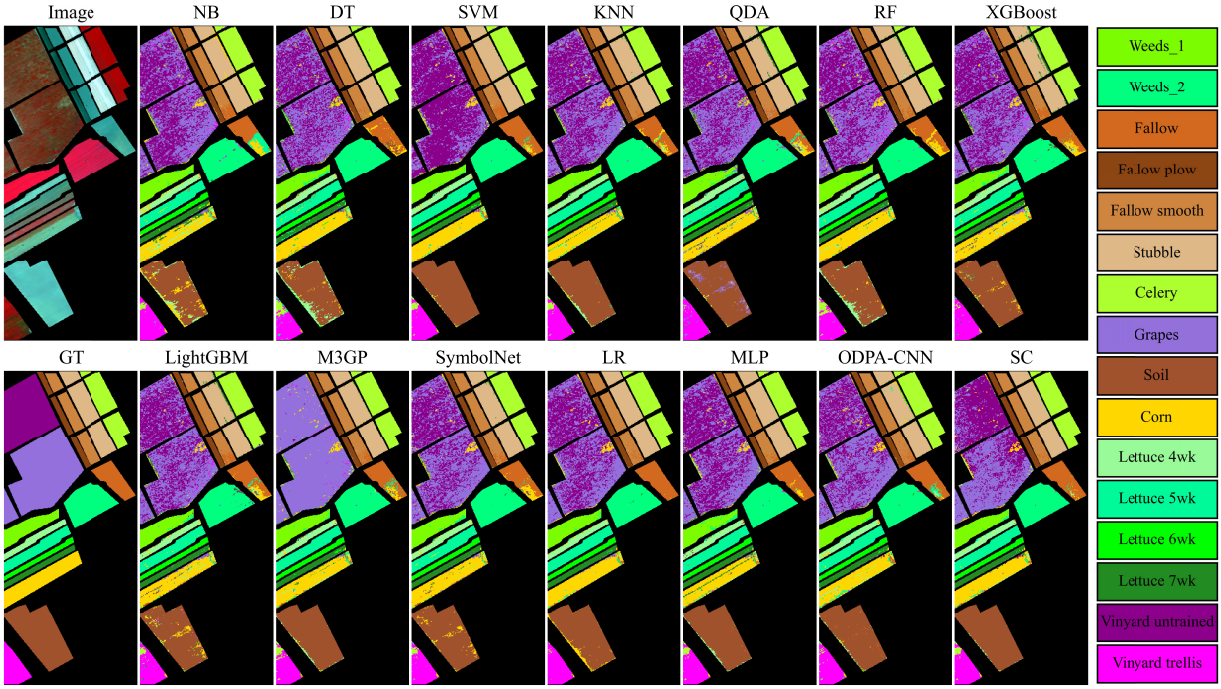


Fig. 10. Classification maps for the Salinas dataset obtained by different methods.

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class-specific expressions, so scalability should be assessed for settings with many classes. Future work will improve search efficiency, introduce stronger physical constraints, and systematically evaluate the domain interpretation of the discovered models.

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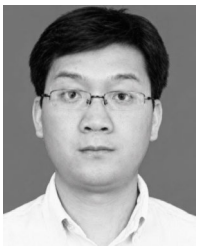
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